

A Temporal Lunch-Contact Model for Gossip Spreading in Workplace Networks

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1 Introduction

Information spreading is one of the most intuitive examples of a dynamical process on a network. In a workplace, rumours, informal warnings, and gossip do not spread in a homogeneous population, but through a structured system of social relations and face-to-face encounters. Some employees are central and well connected, others are peripheral, and some individuals may act as bridges between otherwise weakly connected groups.

The aim of this study is to develop a transparent and usable computational model for exploring such processes. Rather than proposing a universal theory of gossip, the goal is to put a flexible simulation tool into the reader's hands: a model in which different workplace scenarios can be generated, visualized, modified, and compared. The model is intentionally simple, but it contains enough structure to produce non-trivial behaviour and to support interpretable conclusions.

The proposed model combines two components. The first is a stable acquaintance network, where nodes represent employees and edges represent social familiarity. This layer shapes who tends to sit together and whether an active spreader feels socially comfortable enough to start gossiping at a table. The second component is a temporal lunch-contact process, where employees are assigned to circular lunch tables on each simulated day. Once gossip is mentioned at a table, any susceptible tablemate may hear it, with a probability that decreases with seating distance.

This separation between stable social structure and temporary contact opportunities makes it possible to ask questions that would be difficult to study in a purely static network model. For example, how different is gossip spreading on the static acquaintance graph compared with spreading through daily lunch contacts? Does the initial source matter? Do larger lunch tables accelerate diffusion when the total number of seats is fixed? Does a preference for sitting near acquaintances help the rumour spread, or does it trap the information inside local social circles? What changes if informed employees eventually lose interest and stop spreading the gossip?

The contribution of this work is therefore a graph-based stochastic toy model and a set of example analyses demonstrating its use. The model is implemented in Python, visualized

through both the acquaintance graph and the lunch-contact configurations, and evaluated under several parameter settings. The source code is available online.¹ The emphasis is on modelling as an exploratory tool: by changing a small number of interpretable parameters, one can observe different spreading scenarios and draw conclusions about the role of network structure, temporal contacts, seating arrangements, source position, and rumour stifling.

2 Related works and motivation

The proposed lunch-contact model is related to several ideas from network science, epidemic modelling, temporal networks, and workplace gossip research. A general modelling motivation comes from the compartmental view of spreading processes described by Barabási (2016). In epidemic models, individuals are often classified into states such as susceptible, infectious, and recovered. A similar state-based abstraction can be used for rumour spreading, but with a different interpretation: susceptible employees S have not yet heard the gossip, informed employees I know it and may actively spread it, and inactive employees R still know the gossip but no longer transmit it. The analogy is therefore structural rather than biological.

Classical rumour-spreading models provide a more specific background for the epidemic analogy. Nekovee et al. (2007) describe rumours as an “infection of the mind” and emphasize that their spreading resembles epidemic diffusion, while also having its own social rules. They combine the SIR framework with the Maki–Thompson model, in which individuals are classified as ignorant, spreaders, or stiflers. In this model, rumour transmission is directional and rule-based: a spreader initiates a contact and may turn an ignorant individual into a new spreader, but may also become a stifler after contacting other spreaders or stiflers.

Similarly, Lind et al. (2007) study gossip as information about a specific target node in a social network. They introduce two useful quantities: the spread factor, which measures the eventual reach of the gossip among the target person’s acquaintances, and the spreading time, which measures how quickly this reach is achieved. Their results show that the number and arrangement of friendship connections strongly shape gossip propagation. This motivates the present work’s focus on network position of the initial source.

Another important motivation is the distinction between static and temporal contact networks. Many simple spreading models assume that the network is fixed and that all edges are available for transmission at each time step. In real social systems, however, contacts occur at particular moments. Perra et al. (2012) introduced activity-driven temporal networks to describe time-varying contacts, while Barrat et al. (2013) emphasized that empirical face-to-face contact networks contain temporal patterns that are lost in static aggregated representations. This motivates the central comparison in the present study: gossip spreading on the static acquaintance graph versus spreading through daily lunch-contact networks.

The role of weak ties and repeated local contacts is also important. Csermely (2009) discusses how weak links can connect otherwise separated parts of social systems and support

¹GitHub repository: <https://github.com/NemethFloraBoroka/lunch-gossip-model>

communication between groups. In a related temporal-network context, Karsai et al. (2014) showed that strong, recurrent ties may sometimes slow down diffusion by keeping information trapped within repeatedly interacting local neighbourhoods. This idea is directly connected to the friend-seating-bias parameter in the present model. If employees preferentially sit near acquaintances, local transmission may become easier, but global spreading may become slower if the same social circles repeatedly interact. Thus, the model can be used to test whether socially biased seating accelerates or restricts gossip diffusion.

The lunch table can be interpreted as a temporary small-group interaction. As Iacopini et al. (2019) emphasize, pairwise interactions are often not sufficient to describe social contagion, because social influence may also arise in group settings. The present model follows this idea in a simple way: lunch tables act as small-group contexts for gossip spreading, where one active spreader can expose several tablemates at once, with transmission probabilities depending on circular seating distance.

Workplace gossip research provides the organizational interpretation of the model. Ellwardt et al. (2012) studied gossip at work from a social network perspective and showed that gossip patterns depend on social position and group structure. This supports the assumption that gossip does not spread uniformly across all employees. Instead, network position, local communities, and opportunities for informal interaction all matter. In the present study, these ideas are represented through the acquaintance graph, the stochastic lunch seating mechanism, and the comparison of different initial source nodes.

Finally, Csermely et al. (2013) provide a broader methodological motivation for using network-based descriptions of complex systems. Although their review focuses on molecular networks and drug discovery, their general argument is relevant here: network models can provide a useful middle ground between overly reductionist descriptions and unrealistically detailed models that attempt to include everything. They also emphasize the value of network visualization as a way to understand complex systems. These principles also motivate the present lunch-contact model: it is deliberately simple, but its components can be varied, simulated, and visualized to explore different workplace gossip scenarios.

3 Model description

The model developed in this work is a stochastic toy model of gossip spreading in a workplace lunchroom. Its purpose is not to reproduce all psychological and organizational aspects of gossip, but to provide a transparent simulation framework in which different acquaintance network and lunchroom scenarios can be generated, visualized, and compared.

In network terms, employees are represented as nodes, and acquaintance relations are represented as edges. The model also uses basic structural concepts from network science. Employees with many acquaintances can be interpreted as local hubs, while employees with high betweenness centrality may act as bridges between otherwise weakly connected parts of the workplace network.

The central modelling idea is that workplace gossip depends on both stable social relations and temporary face-to-face encounters. The acquaintance network does not directly determine who is allowed to hear the gossip. Instead, it influences who tends to sit together at lunch and whether an active spreader feels socially comfortable enough to start gossiping at a table. Once the gossip is mentioned, however, any susceptible tablemate may hear it, with a probability that decreases with seating distance.

Thus, the model separates three mechanisms: the stable acquaintance graph, the daily lunch seating process, and the stochastic spreading rule at lunch tables. In this sense, gossip spreading is treated as a dynamical process on top of a structured social network, modified by the temporal contact opportunities generated by lunch seating.

3.1 The workplace acquaintance network

Let $G_F = (V, E_F)$ denote the workplace acquaintance network. The set of nodes V represents employees, and an edge $(i, j) \in E_F$ means that employees i and j know each other. The number of employees is denoted by $N = |V|$.

The implementation allows several graph types, including Erdős–Rényi, Barabási–Albert, Watts–Strogatz, and stochastic block model graphs. In the simulations of this study, the stochastic block model is used because it can represent a workplace with several social groups or departments. These groups create modularity in the acquaintance network, but they do not represent physically separated company sites.

In the stochastic block model, employees are assigned to K groups,

$$V = V_1 \cup V_2 \cup \dots \cup V_K.$$

The probability of an acquaintance edge:

$$P((i, j) \in E_F) = \begin{cases} p_{\text{in}}, & \text{if } i \text{ and } j \text{ are in the same group,} \\ p_{\text{out}}, & \text{if } i \text{ and } j \text{ are in different groups.} \end{cases}$$

Usually, $p_{\text{in}} > p_{\text{out}}$, which creates dense local social circles together with weaker connections between groups.

3.2 Daily lunch seating

At each discrete time step $t = 1, 2, \dots$, one lunch event takes place. Each employee attends lunch with probability p_{attend} . The attending employees are then assigned to circular lunch tables.

The lunchroom is controlled by two capacity parameters. The maximum number of people at one table is denoted by s , while the preferred number of available tables is denoted by n_{tables} . Together, these parameters determine the total seating capacity,

$$C = s \cdot n_{\text{tables}}.$$

If n_{tables} is not specified, the model uses the minimum number of tables required to seat the attending employees. If it is specified, the model uses this fixed number whenever possible. This enables comparison of lunchroom layouts with the same total number of chairs but different table sizes.

The seating process is random, but biased by the acquaintance network. Employees are more likely to join a table where socially close employees are already seated. Direct acquaintances contribute a full seating attraction, while two-step acquaintances contribute a reduced attraction. This is represented by the seating proximity score

$$h(i, j) = \begin{cases} 1, & \text{if } d_F(i, j) = 1, \\ \lambda, & \text{if } d_F(i, j) = 2, \\ 0, & \text{otherwise,} \end{cases}$$

where $d_F(i, j)$ is the shortest-path distance in the acquaintance graph and $0 \leq \lambda \leq 1$. In the implementation, λ is the seating factor. It affects where people sit, not whether they are allowed to hear the gossip.

If an employee i is considered for a table T , the table receives a social seating score

$$H(i, T) = \sum_{j \in T} h(i, j).$$

The sampling weight for seating i at that table is then proportional to

$$1 + bH(i, T),$$

where $b \geq 0$ is the friend seating bias. When $b = 0$, seating is approximately random. Larger values of b create more socially clustered lunch tables.

3.3 Lunch-table exposure and transmission

For each day t , the seating process generates a temporary lunch-contact structure,

$$G_L(t) = (V_t, E_L(t)),$$

where $V_t \subseteq V$ is the set of employees attending lunch, and $E_L(t)$ contains pairs of employees seated at the same table. However, the model does not treat all tablemates as equally close. Because tables are circular, the distance between two employees sitting at the same table is measured by circular seat distance.

If employees i and j sit at positions pos_i and pos_j around a table of size m , their circular distance is

$$d_{\text{seat}}(i, j) = \min(|pos_i - pos_j|, m - |pos_i - pos_j|).$$

Employees sitting next to each other have $d_{\text{seat}} = 1$. Larger values mean that they are

farther apart around the table. The effect of seating distance on exposure is

$$g(d_{\text{seat}}(i, j)) = \exp[-\alpha(d_{\text{seat}}(i, j) - 1)],$$

where $\alpha \geq 0$ controls how quickly exposure probability decreases with distance.

A key rule of the model is that an active spreader does not automatically start gossiping at every lunch table. An informed employee i starts gossiping only if at least one direct acquaintance is present at the same table. This represents the idea that employees may not spread a sensitive rumour among complete strangers. Once the gossip is mentioned, however, every susceptible tablemate j may hear it. The probability of transmission is

$$p_{i \rightarrow j}(t) = \beta \cdot g(d_{\text{seat}}(i, j)),$$

or equivalently,

$$p_{i \rightarrow j}(t) = \beta \cdot \exp[-\alpha(d_{\text{seat}}(i, j) - 1)].$$

Here, β is the baseline probability of transmitting the gossip. This rule separates the social trigger for starting gossip from the exposure process at the table: acquaintances make gossiping more likely to begin, but non-acquaintances can still overhear it once it is being discussed.

3.4 Gossip states

The simplest version of the model uses two states. Each employee is either susceptible, S , meaning that they have not yet heard the gossip, or informed, I , meaning that they know the gossip and can transmit it further. This corresponds to an SI spreading process.

At the beginning of the simulation, one employee is selected as the initial source. The source can be chosen randomly, or according to a network property such as degree or betweenness centrality:

$$X_i(0) = \begin{cases} I, & \text{if } i = i_0, \\ S, & \text{otherwise,} \end{cases}$$

where i_0 is the initial source.

The model can also be run in an SIR-like rumour mode. In this version, an informed employee may become inactive. Inactive employees still know the gossip, but no longer transmit it. The states are then

$$S \rightarrow I \rightarrow R,$$

where R denotes an informed but inactive employee. After each day, an active spreader becomes inactive with probability ρ . This represents the loss of novelty or interest in the rumour.

3.5 Main parameters of the model

The model contains a small number of adjustable parameters. These parameters are not meant to be exact empirical measurements. Instead, they make it possible to explore hypothetical workplace and lunchroom scenarios in a controlled way.

Parameter	Meaning	Expected effect
N	Number of employees	Larger networks may require more time for widespread diffusion.
$\bar{k}$	Target average degree, used only when edge probabilities are not fixed	Higher values create denser automatically generated acquaintance graphs.
K	Number of social groups in the stochastic block model	More or stronger groups can make the acquaintance network more modular.
p_{in}	Within-group edge probability	Higher values create denser local social circles.
p_{out}	Between-group edge probability	Higher values create more bridges between groups.
s	Maximum lunch table size	Larger tables create more potential table-level exposure.
n_{tables}	Preferred number of lunch tables	Together with s , determines total seating capacity.
β	Baseline transmission probability	Higher values increase spreading speed once gossip is mentioned.
λ	Weight of two-step acquaintances in seating	Higher values make indirect acquaintances more relevant for seating.
α	Seating distance decay parameter	Higher values localize exposure around nearby seats.
b	Friend seating bias	Higher values make acquaintances more likely to sit together and trigger gossiping.
p_{attend}	Daily lunch attendance probability	Lower values reduce daily exposure opportunities.
ρ	Inactivation probability in SIR mode	Higher values may stop spreading before high coverage is reached.

Table 1: Main parameters of the lunch-contact gossip model.

3.6 Simulation outputs and research questions

Because the model is stochastic, a single simulation run is not sufficient to characterize its behaviour. The same parameter setting may lead to different outcomes depending on the generated acquaintance graph, the initial source, the attendance pattern, and the lunch seating. Therefore, each experiment is repeated many times, and the results are summarized by averages, standard errors, distributions, and spreading curves.

The main time-dependent output is the informed fraction,

$$F(t) = \frac{1}{N} \sum_{i \in V} \mathbf{1}\{X_i(t) \in \{I, R\}\},$$

where $F(t)$ is the fraction of employees who know the gossip by day t . In SI mode, informed employees are active spreaders. In SIR mode, both active and inactive informed employees are counted as informed, but only active I -state employees can transmit the gossip further.

A useful summary measure is the time required to inform 80% of employees,

$$T_{80} = \min\{t : F(t) \geq 0.8\}.$$

This measure is more robust than the time to full coverage, because full coverage may depend strongly on the last few peripheral employees or may not occur within the simulation horizon.

In addition to numerical summaries, the model is visually interpretable. The acquaintance graph can be plotted to show the stable social structure, the initial source, and its local network neighbourhood. Individual lunch configurations can also be visualized as circular tables, making it clear how daily exposure opportunities are generated. For aggregated results, the model produces degree distributions, average spreading curves, T_{80} boxplots, and active-spreader curves in the SIR extension. These visual outputs make the model useful not only for simulation, but also for understanding how network structure, seating dynamics, and rumour activity jointly shape the spreading process.

4 Results

The purpose of the simulations was not to fit the model to empirical workplace data, but to demonstrate how the proposed lunch-contact framework can be used as an exploratory tool. Unless otherwise stated, the simulations used a workplace with $N = 50$ employees, generated by a stochastic block model with $K = 3$ social groups, $p_{\text{in}} = 0.30$, and $p_{\text{out}} = 0.03$. The baseline lunchroom contained $n_{\text{tables}} = 12$ circular tables with maximum table size $s = 6$. The main transmission parameters were $\beta = 0.30$, $\lambda = 0.25$, $\alpha = 0.70$, and $b = 3.0$. Each stochastic experiment was repeated 100 times over a maximum of 60 simulated days.

Figure 1a shows one example of the generated acquaintance network. The nodes represent employees, and the edges represent social familiarity. The stochastic block structure creates local groups while still allowing some connections between groups. The highlighted node and its

neighbourhood illustrate how the initial source and its local social environment can be inspected visually.

Figure 1b shows the corresponding lunch-contact configuration on the first simulated day. Employees are placed at circular tables, and gossip transmission can occur only between tablemates. This visualization alongside the acquaintance network illustrates the key modelling assumption of this study: the stable acquaintance graph shapes seating and determines whether gossip is socially triggered, while the daily lunch seating determines who is exposed once the gossip is mentioned.

Before analysing the spreading dynamics, I characterized the generated acquaintance networks by their degree distribution. Figure 1c shows the average degree distribution across repeated stochastic block model realizations. Averaging over many generated networks avoids drawing conclusions from the accidental structure of a single random graph. This step provides better understanding of subsequent experiments.

The first spreading experiment compared two mechanisms: spreading only on the static acquaintance graph and spreading through the temporal lunch-contact model. In the static baseline, gossip can spread directly through acquaintance relations at every time step. In the lunch-contact model, the acquaintance graph has a different role: it shapes who tends to sit together and determines whether an active spreader starts gossiping at a table. Specifically, an active spreader only starts talking if at least one direct acquaintance is present at the same table. Once the gossip is mentioned, however, any susceptible tablemate may hear it, with a probability that decreases with circular seating distance. Figure 1d shows that the static graph produces faster spreading, while the temporal lunch-contact model delays diffusion. This confirms that the lunchroom mechanism is not a minor detail: limiting transmission to temporary face-to-face encounters, and requiring a socially plausible trigger for gossiping, substantially changes the spreading curve.

The next experiment tested whether the initial source of the gossip matters. Four source-selection strategies were compared: a random employee, the highest-degree employee, the lowest-degree employee, and the employee with the highest betweenness centrality. The outcome was summarized by T_{80} , the first day on which 80% of employees had heard the gossip. As shown in Figure 1e, peripheral starting nodes tend to produce slower spreading than more central starting nodes. This illustrates how the same model can be used to compare different network positions: degree captures local popularity, while betweenness captures a more bridge-like role between parts of the network.

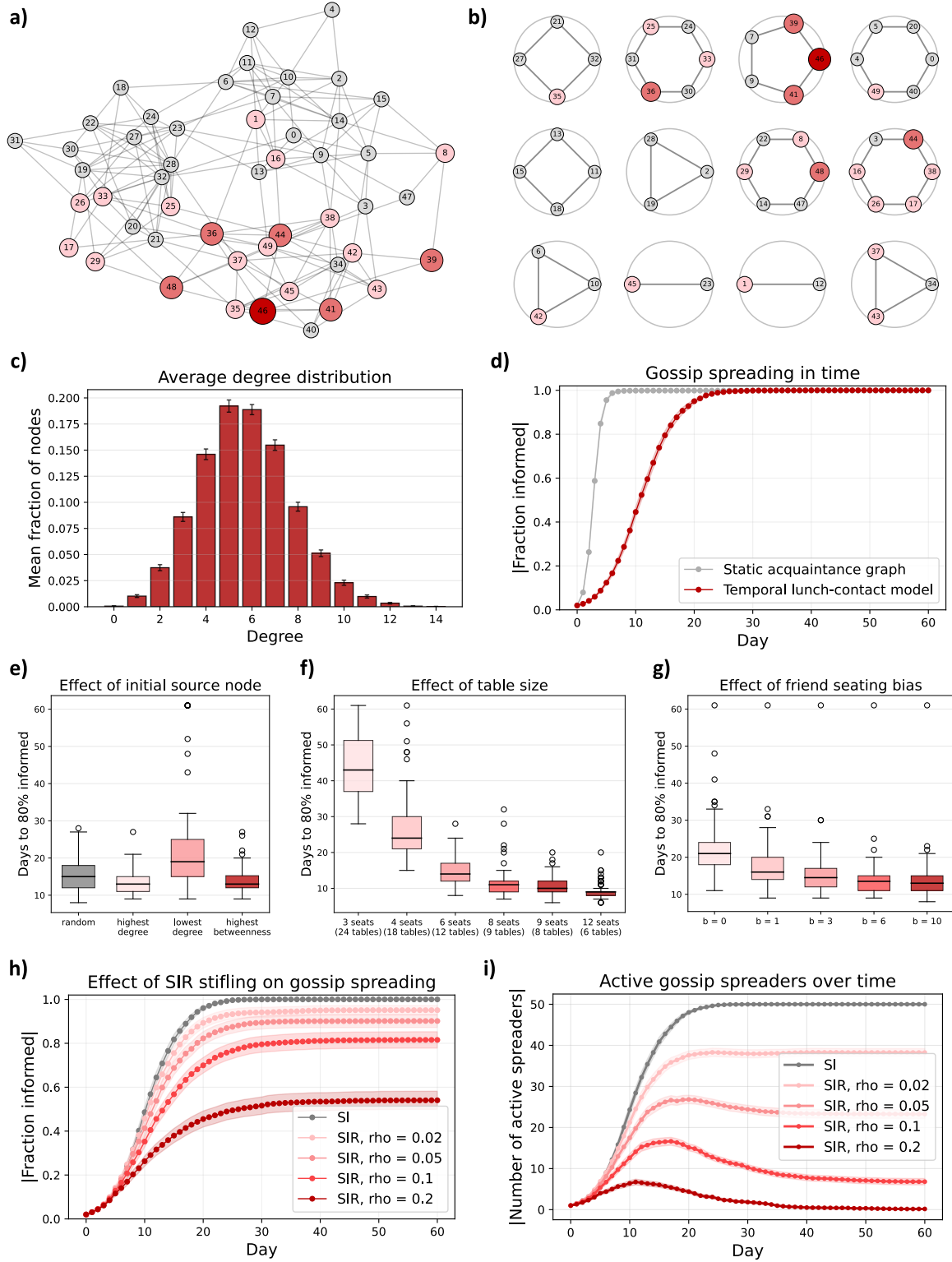


Figure 1: Summary of the temporal lunch-contact gossip model. (a) Example acquaintance network generated by the stochastic block model, with informed individuals highlighted. (b) Example daily lunch seating configuration, where employees are assigned to circular tables and transmission can occur after a socially triggered gossip event at the table. (c) Average degree distribution of the generated acquaintance networks across repeated simulations. (d) Comparison of gossip spreading on the static acquaintance graph and in the temporal lunch-contact model. (e) Effect of the initial source node on the time required to inform 80% of employees. (f) Effect of table size while keeping the total number of cafeteria seats fixed. (g) Effect of friend seating bias, representing the tendency to sit near acquaintances. (h) Effect of the SIR stifling mechanism on the fraction of informed employees. (i) Average number of active gossip spreaders over time in the SIR extension. Together, the panels show that gossip diffusion depends not only on the underlying acquaintance network, but also on temporal lunch contacts, seating structure, source position, and the loss of interest in further spreading.

I then examined the effect of lunch table size while keeping the total number of cafeteria seats fixed. The tested table sizes were $s \in \{3, 4, 6, 8, 9, 12\}$, with the number of tables adjusted so that the total capacity remained 72 seats in every condition. Therefore, the experiment did not change the total lunchroom capacity, only its organization into many small tables or fewer large tables. Figure 1f shows that larger tables generally reduce the time needed to reach 80% informed. This suggests that grouping employees into larger temporary contact groups can accelerate gossip diffusion, even though seating distance still reduces the probability of transmission between distant tablemates.

The following experiment varied the friend seating bias parameter b , which controls how strongly employees prefer to sit at tables where socially close employees are already present. The tested values were $b \in \{0, 1, 3, 6, 10\}$. In the model, this parameter is especially important because the acquaintance network affects the process in two ways. First, it shapes the seating arrangement: higher values of b make acquaintances more likely to sit at the same table. Second, an active spreader only starts gossiping if at least one direct acquaintance is present at the table. Once the gossip is mentioned, however, any susceptible tablemate may hear it. Therefore, socially biased seating increases the chance that the social trigger for gossiping is present, while also creating exposure opportunities for other tablemates. Figure 1g shows that stronger friend seating bias decreased T_{80} , indicating faster spreading. Under these parameter settings, the benefit of bringing acquaintances together dominated any possible reduction in global mixing.

Finally, I extended the model from an SI process to an SIR-like rumour process. In the SI version, informed employees continue spreading indefinitely. In the SIR version, active spreaders may become inactive with probability ρ after each day; inactive employees still know the gossip, but no longer transmit it. The tested values were $\rho \in \{0.00, 0.02, 0.05, 0.10, 0.20\}$, where $\rho = 0$ corresponds to the SI baseline. Figure 1h shows that increasing ρ slows the increase of the informed fraction and can prevent the gossip from reaching the whole workplace within the simulation horizon.

The SIR extension is further illustrated in Figure 1i, which shows the average number of active spreaders over time. In the SI case, informed employees remain active, so the number of spreaders can remain high once the gossip has reached many employees. In contrast, in the SIR cases, active spreaders eventually disappear as individuals lose interest in transmitting the rumour. This panel shows that the informed fraction alone does not fully describe the dynamics: two simulations may have similar final awareness, but very different numbers of employees who are still actively spreading the gossip.

Overall, the experiments show that the proposed model produces interpretable differences between spreading scenarios.

5 Discussion

The aim of this study was not to build a fully realistic model of workplace gossip, but to develop a transparent computational toy model that can be used for systematic experimentation. The results show why such a model is useful. Once the basic mechanisms are implemented, the same framework can be used to generate many different workplace scenarios and to compare them under controlled conditions. Instead of making a single prediction, the model allows the user to ask “what if” questions: What if the gossip starts from a central employee? What if lunch tables are larger? What if people prefer to sit near acquaintances? What if spreaders eventually lose interest?

This is the main advantage of a toy model. It deliberately removes many real-world details, but keeps a small number of interpretable mechanisms. In the present model, these mechanisms are the acquaintance network, the daily lunch seating process, the social trigger for starting gossip at a table, distance-dependent exposure, and optional SIR-like stifling. Because these components are explicit, they can be changed one at a time. This makes the model useful not as a predictive tool, but as a way to reason about how network structure and temporal contact opportunities may shape information spreading.

The simulations also illustrate that the same underlying acquaintance network can play several roles. It can determine who is central or peripheral, influence who tends to sit together, and decide whether an active spreader feels socially comfortable enough to start gossiping at a table. However, once the gossip is mentioned, the model allows any susceptible tablemate to hear it. This distinction is important because it separates the social conditions that make gossiping likely from the exposure process through which information may reach unintended listeners.

Several extensions of the model would be possible. One natural direction would be to introduce multiple company sites or separated lunchrooms. In such a version, employees would usually have lunch only with colleagues from the same site, while rare cross-site lunches or inter-site visitors could create occasional bridges between otherwise separated contact environments. This would make it possible to study whether stable inter-site acquaintance ties or rare physical meetings are more important for organization-wide gossip spreading.

Another extension would be to add memory to the lunch seating process. In the current model, lunch tables are regenerated independently on each simulated day, although the acquaintance network biases the seating. A more realistic version could include seating persistence, where employees tend to return to previous lunch partners or habitual tables. This could strengthen repeated local interactions and might slow down global diffusion if the same groups repeatedly eat together.

The SIR mechanism could also be refined. In the present version, active spreaders become inactive with a fixed daily probability, representing loss of interest or novelty. A more detailed model could make stifling depend on context: for example, employees might stop spreading the rumour once most of their close contacts already know it, or after repeatedly failing to reach new listeners. Such extensions would make the model more realistic, but they would also

introduce additional assumptions and parameters.

Overall, the proposed lunch-contact model demonstrates the value of having a flexible simulation framework. It provides a concrete object that can be modified, visualized, and tested. This is especially useful in network science, where intuition about spreading processes can be misleading: a small change in contact timing, source position, seating structure, or activity decay can substantially change the observed diffusion pattern. The model should therefore be understood as an exploratory tool for thinking about workplace gossip, rather than as a final description of real organizational behaviour.

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